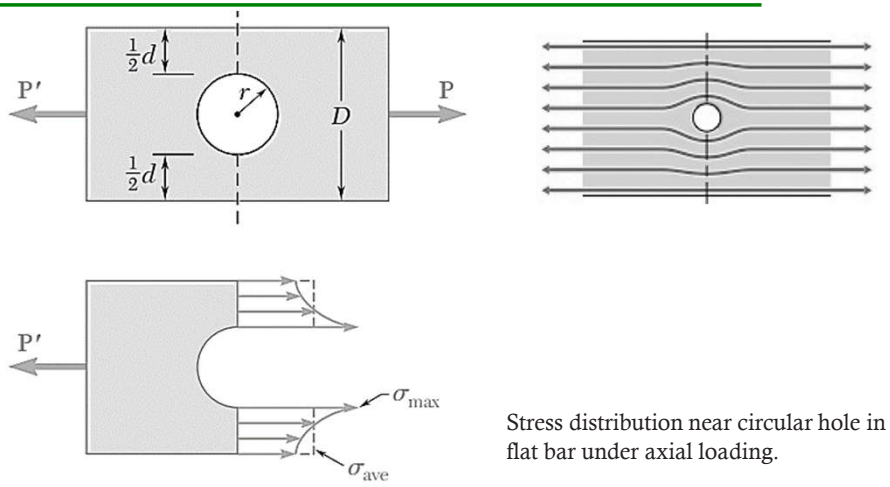
Stress Concentrations

Stress Concentrations

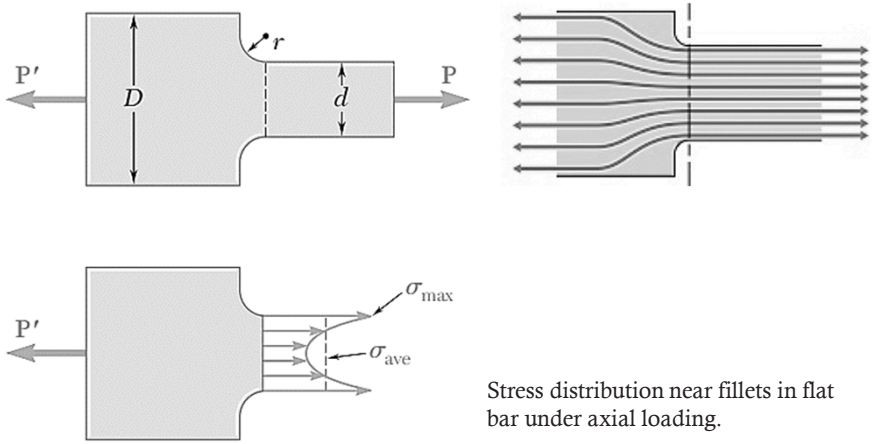
- Recall Saint Venant's Principle.
- When a structural member contains a discontinuity, such as a hole or a sudden change in cross section, high localized stresses can occur near the discontinuities.
- The maximum stress in such members is calculated as

$$\sigma_{\max} = K\sigma_{nom}$$

  - where the nominal stress is calculated with the elementary formula.



Stress distribution near circular hole in flat bar under axial loading.



Stress distribution near fillets in flat bar under axial loading.

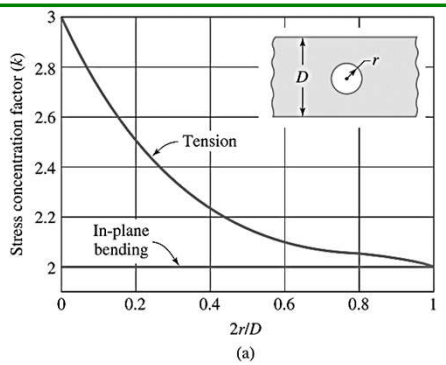
Stress Concentration Factors

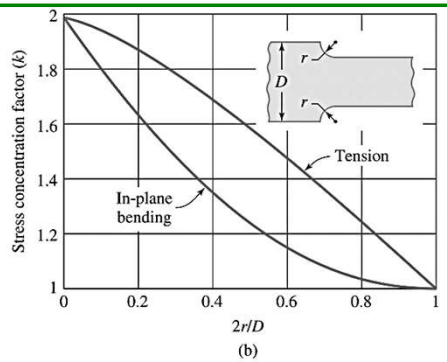
- Values of K can be read from curves available in design handbooks.

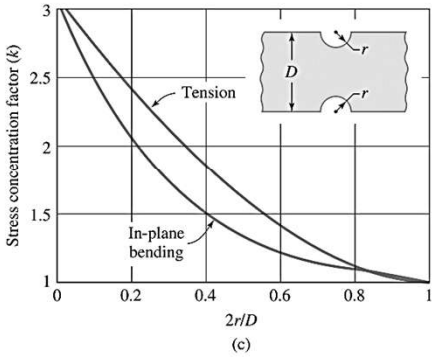
- K (aka K_t) are often used when normal nominal stress is calculated using net cross-sectional area (where applicable).

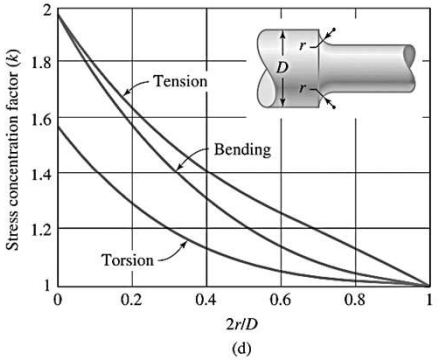
$$\text{So} \quad \sigma_{\max} = K\frac{P}{A_{net}}$$

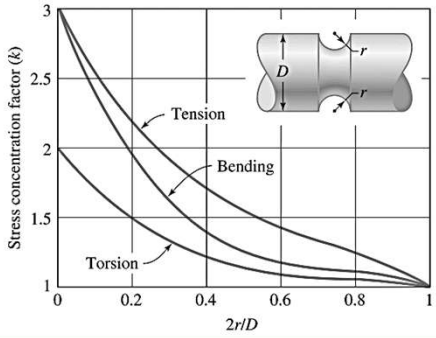
- Relations also exist for calculating these factors. (See Mechanics of Materials by Pytel and Kiusalaas).





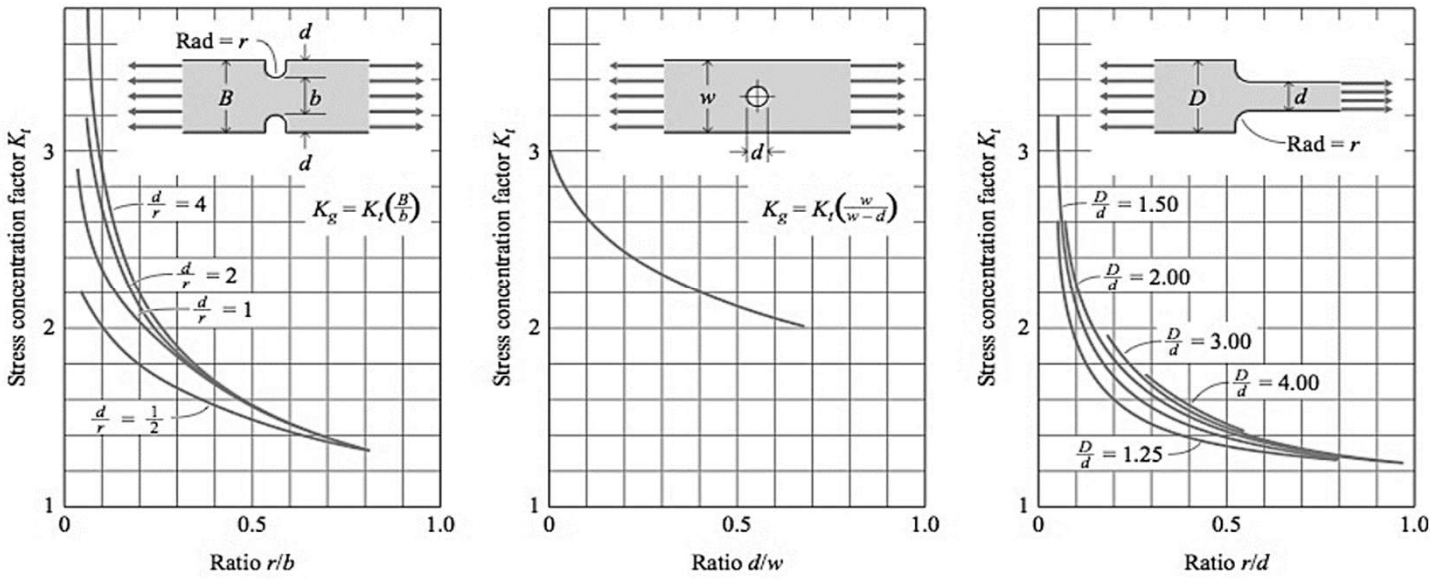


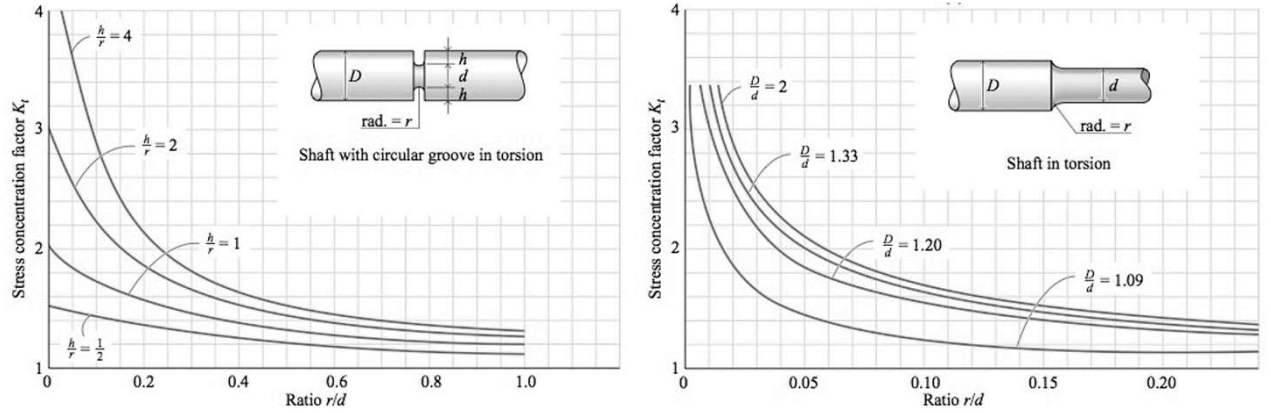




Stress Concentrations

- Appropriate conversions to K_g can be made, where one wants to work with gross area (in nominal stress calculations).





Stress Concentration

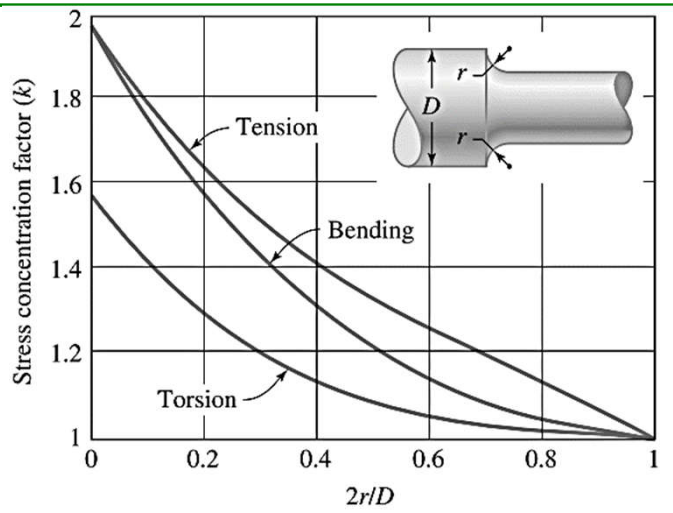
- For Torsional stress,

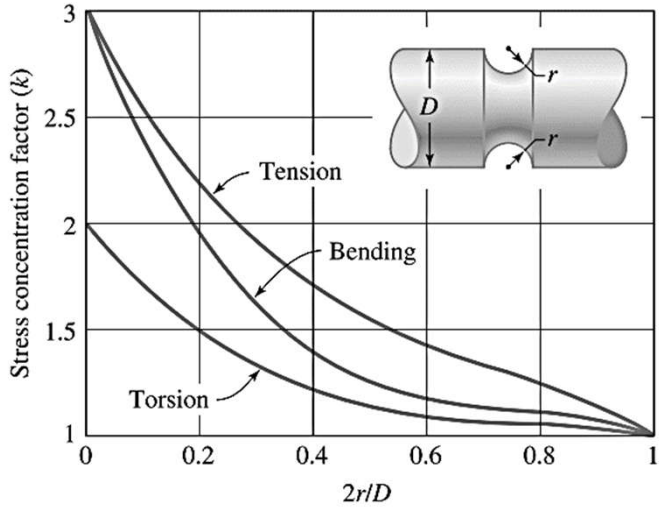
$$\tau_{\max} = K_t \frac{Tr}{J}$$

Note: Where there is a change in diameter, the radius of the smaller shaft is used in calculating the stress

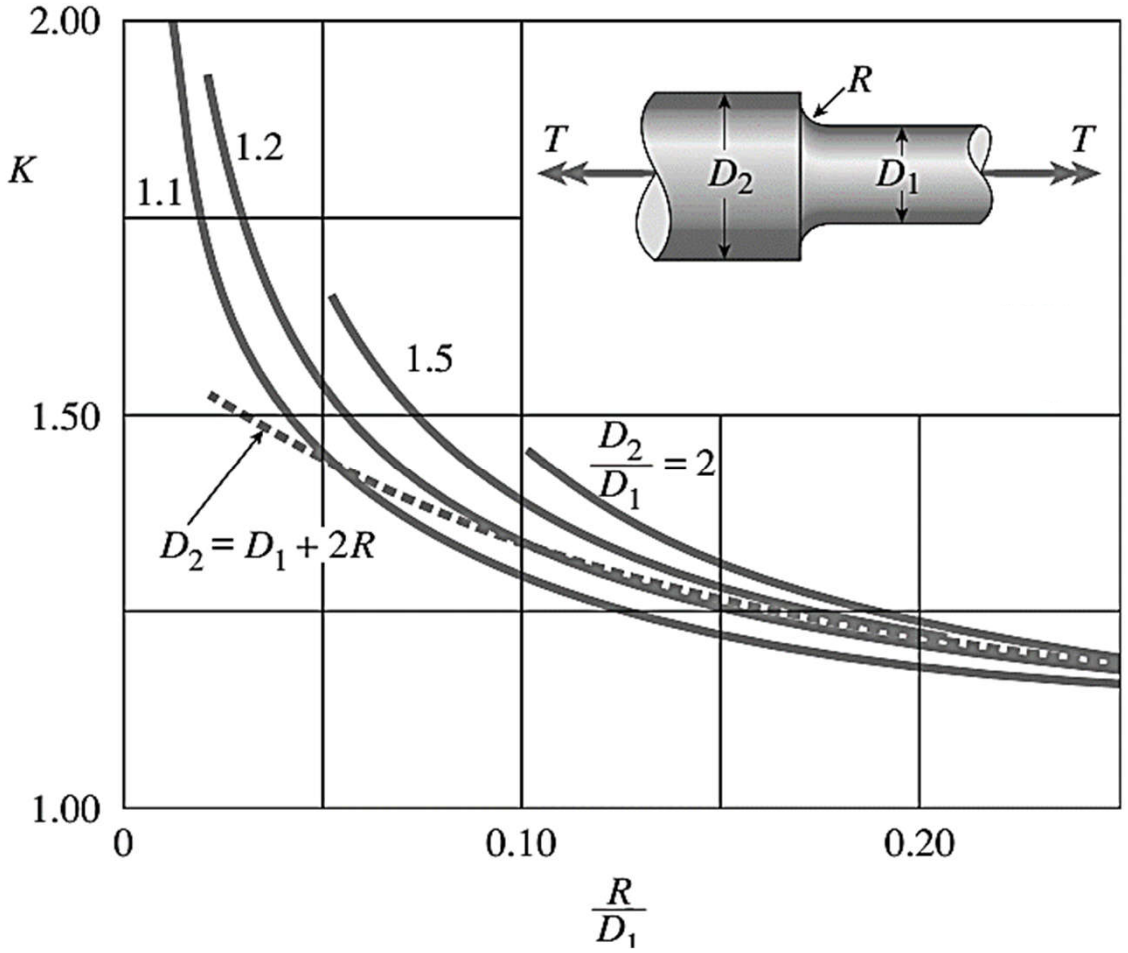
- For bending;

$$\sigma_{\max} = K_t \frac{My}{I}$$





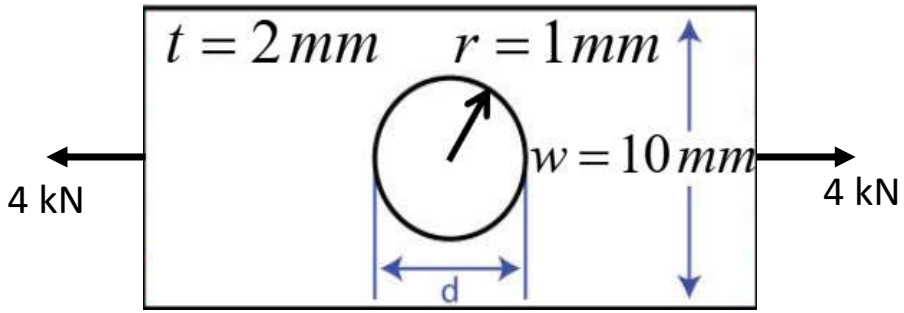
Stress Concentration

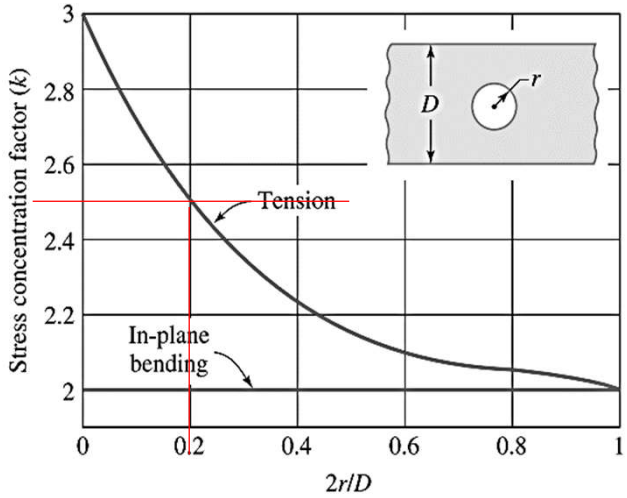


Stress Concentrations

Example

Calculate the stress in a plate of dimensions shown if it is in tension under forces of 4 kN at each side.





$$\frac{2r}{d} = \frac{2(1)}{10} = 0.2$$

From the chart, for 2r/d=0.2,

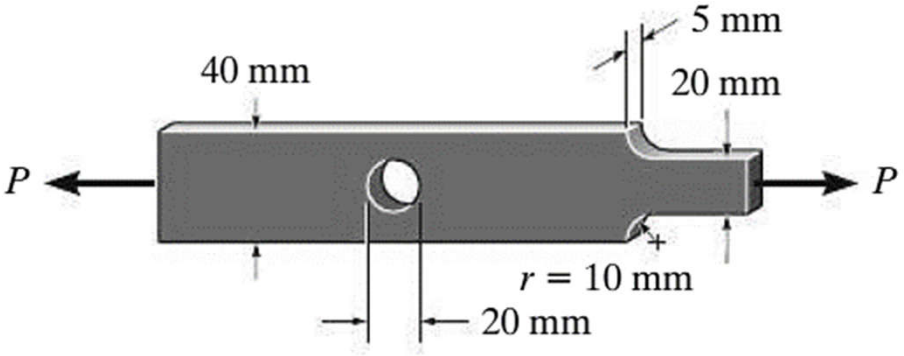
$k \simeq 2.5$

$$\sigma_{max} = k\frac{P}{A_{net}} = \frac{2.5(4000)}{2(10-2)\times 10^{-6}} = 625 \text{ MPa}$$

Stress Concentrations

Example

Calculate the stress in a plate of dimensions shown if it is in tension under forces of P = 8-kN at each side.



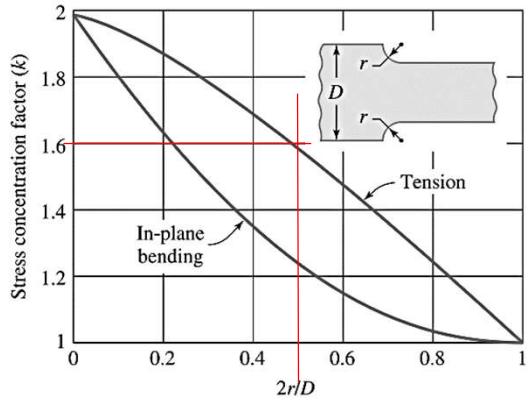
Maximum stress will either be in the fillet or on the boundary of the holes.
At the fillet,

$$\frac{2r}{D} = \frac{2(10)}{40} = 0.5$$

From the chart, for 2r/D=0.5,

$k \simeq 1.6$

$$\sigma_{max} = k\frac{P}{A_{net}} = 1.6\left(\frac{8000}{20\times 5\times 10^{-6}}\right) =$$



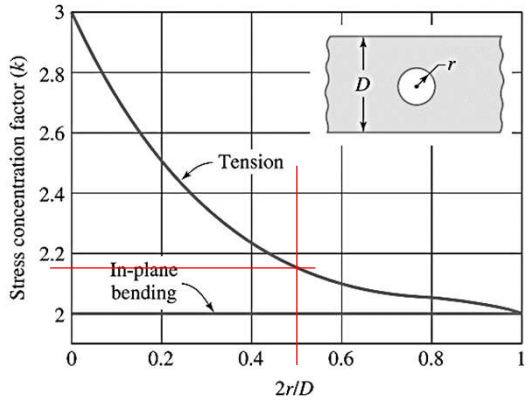
At the hole,

$$\frac{2r}{D} = \frac{20}{40} = 0.5$$

From the chart, for 2r/D=0.5,

$k \simeq 2.15$

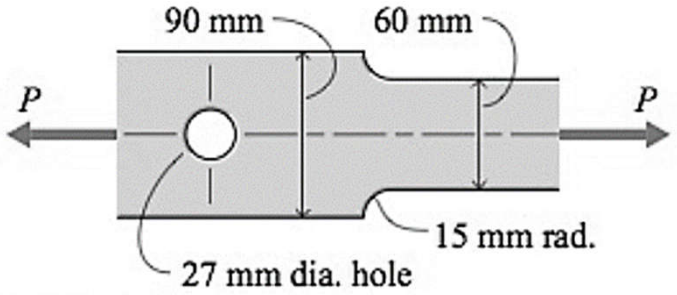
$$\sigma_{max} = k\frac{P}{A_{net}} = 2.15\left(\frac{8000}{(40-20)\times 5\times 10^{-6}}\right) =$$



Stress Concentrations

Example

The machine part shown below is 20 mm thick and is made of 0.4 percent carbon hot-rolled steel. Determine the maximum safe load P if the maximum normal stress is not to exceed 144 MPa.



Maximum stress will either be in the fillet or on the boundary of the holes.

At the fillet,

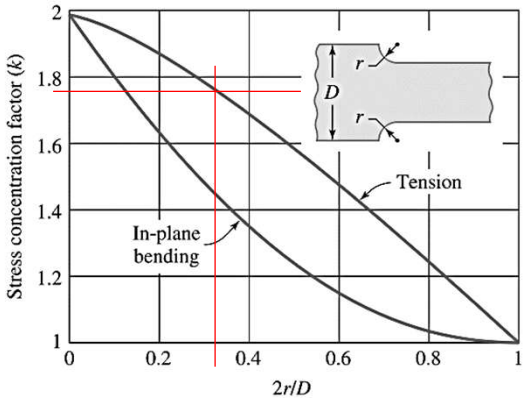
$$\frac{2r}{D} = \frac{2(15)}{90} = 0.333$$

From the chart, for 2r/D=0.33,

$k \simeq 1.75$

$$\sigma_{\max} = k\frac{P}{A_{net}}$$

$$P_{\max} = \frac{\sigma_{\max} A_{net}}{k} = \frac{(144\times10^{6})(20\times60\times10^{-6})}{1.8} = 98.7 \text{ kN}$$



At the hole,

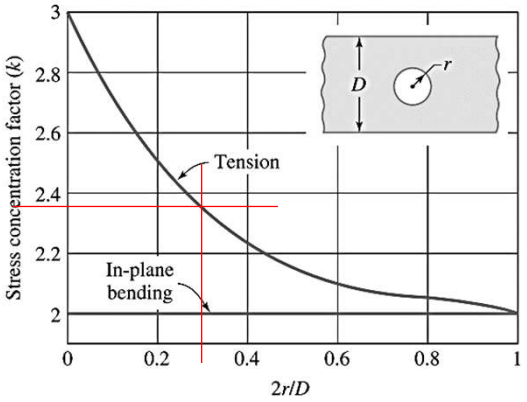
$$\frac{2r}{D} = \frac{27}{90} = 0.3$$

From the chart, for 2r/D=0.3,

$k \simeq 2.35$

$$\sigma_{\max} = k\frac{P}{A_{net}}$$

$$P_{\max} = \frac{\sigma_{\max} A_{net}}{k} = \frac{(144\times10^{6})(20\times(90-27)\times10^{-6})}{2.35} = 78.9 \text{ kN}$$



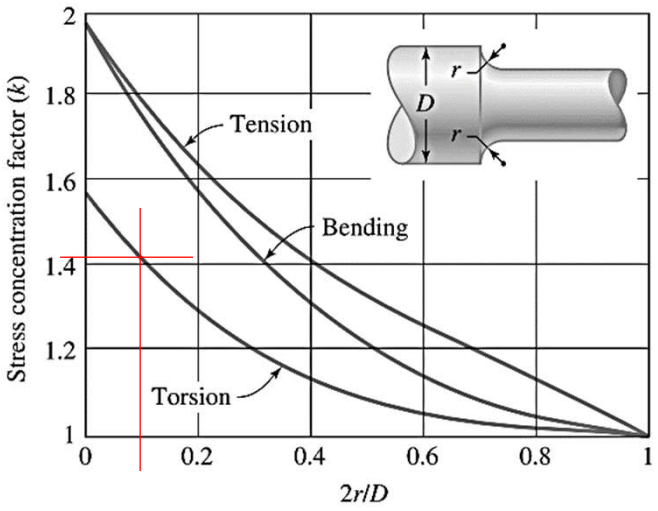
Stress Concentration

Example

A fillet with a radius of **6** mm is used at the junction in a stepped shaft where the diameter is reduced from 135 mm to 100 mm. Determine the maximum shearing stress in the fillet when the shaft is transmitting a torque of 10 kN-m.

$$\frac{2r}{D} = \frac{12}{135} = 0.089$$

$$K \cong 1.42$$



$$\tau_{max} = K_t \frac{Tr}{J}$$

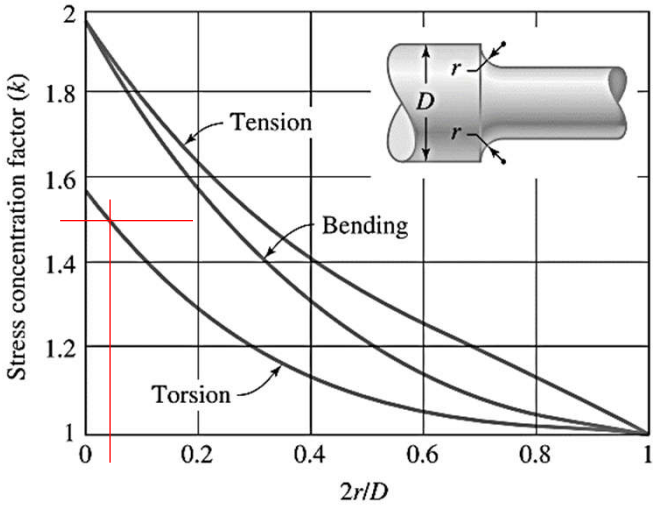
$$J = 9.8175 \times 10^{-6} \text{ m}^4$$

$$\tau_{max} = \frac{1.42(10000)(0.05)}{9.8175 \times 10^{-6}} =$$

Stress Concentrations

Example

A stepped shaft consisting of solid circular segments having diameters D_1 = 2.0 in. and D_2 = 2.4 in. is subjected to torques T at either end. The radius of the fillet is R = **0.05** in. If the allowable shear stress at the stress concentration is 6000 psi, what is the maximum permissible torque?



$$\frac{2r}{D}=\frac{0.1}{2.4}=0.042$$

$$K\cong 1.51$$

$$\tau_{\max}=K_t\frac{Tr}{J}$$

$$J=1.5708\ \text{in}^4$$

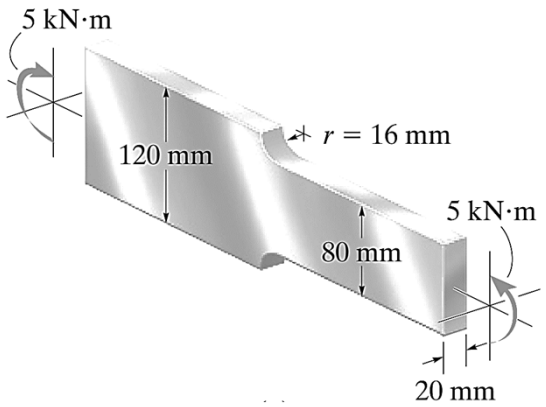
$$6000=\frac{1.51(T)(1)}{1.5708}$$

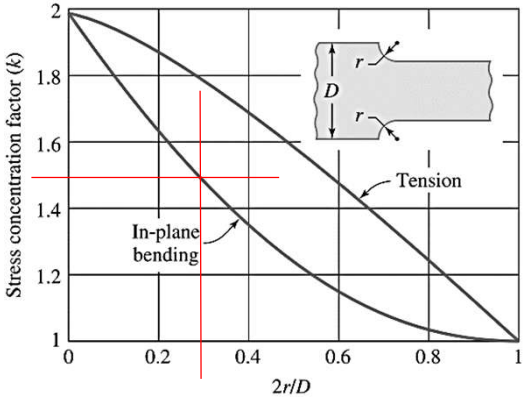
$$T=$$

Stress Concentrations

Example

The transition in the cross-sectional area of the steel bar is achieved using shoulder fillets, as shown below. If the bar is subjected to a bending moment of 5 kNm, determine the maximum normal stress developed in the steel. The yield stress is = 500 MPa





$$\frac{2r}{D} = \frac{2(16)}{120} = 0.3$$

From the chart, for 2r/D=0.3,

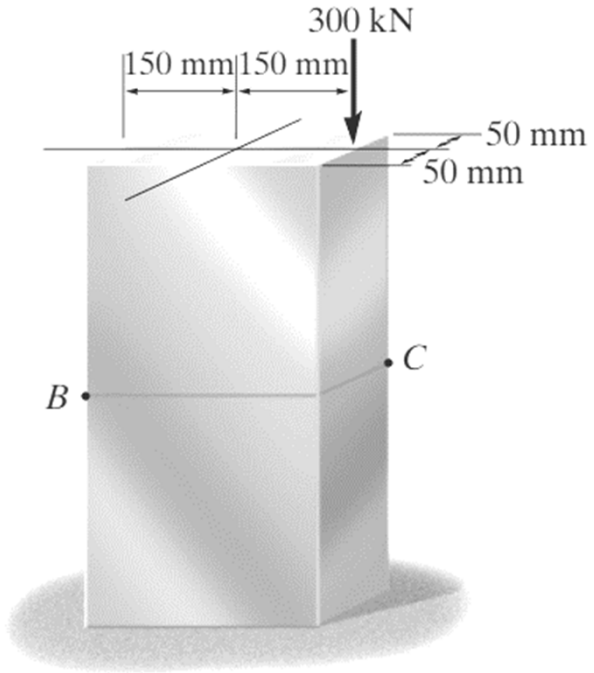
$k \simeq 1.5$

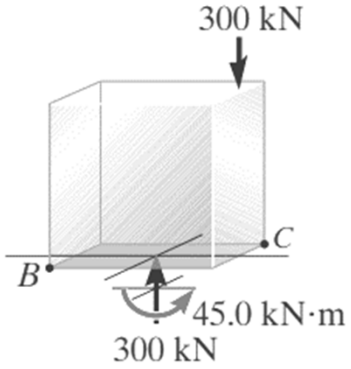
$$\sigma_{max} = k\frac{My_{max}}{I} = 1.5\left(\frac{5000(0.04)(12)}{0.02(0.08^3)}\right) =$$

Combined Loading

Combined Loading

A force of 300 kN is applied to the edge of the member shown below. Neglect the weight of the member and determine the state of stress at points B and C.



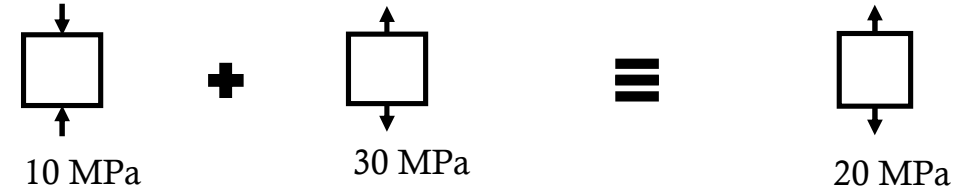


$$\sigma_{axial} = 10 \text{ MPa (C)}$$

$$\sigma_{\max\ (bending)} = 30 \text{ MPa}$$

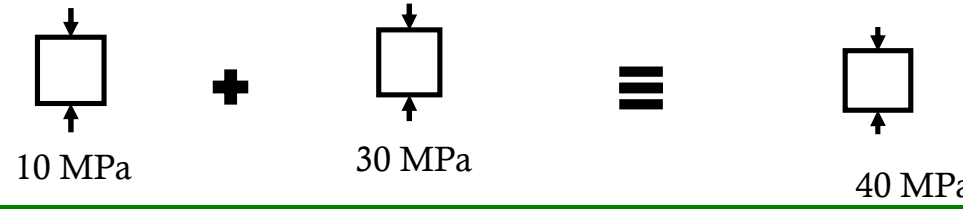
At B,

Axial + Bending ≡



At C,

Axial + Bending ≡

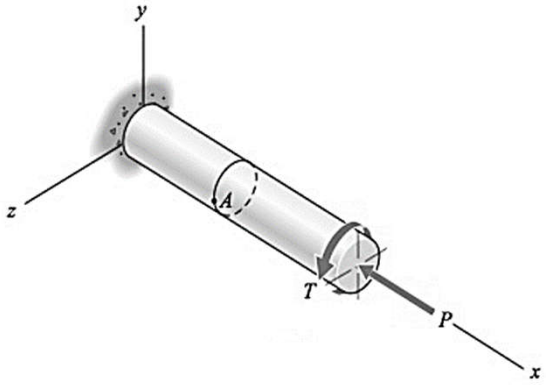


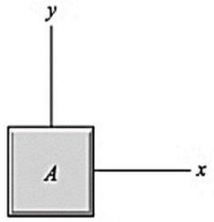
Combined Loading

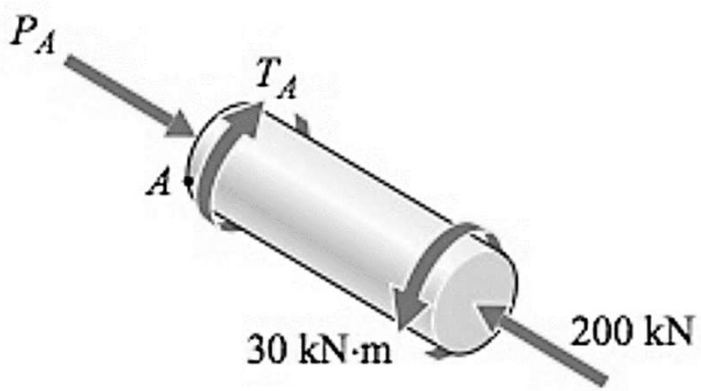
The solid 100-mm-diameter shaft shown below is subjected to an axial compressive force of P = 200 kN and a torque of T = 30 kNm. For point A on the outside surface of the shaft, determine

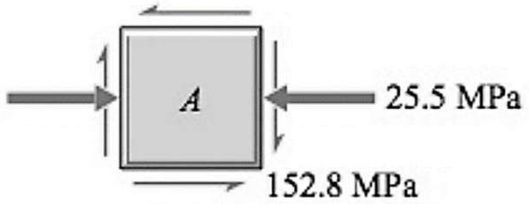
(a) The x- and y-components of stress.

(b) The principal stresses and the maximum shearing stress at the point.









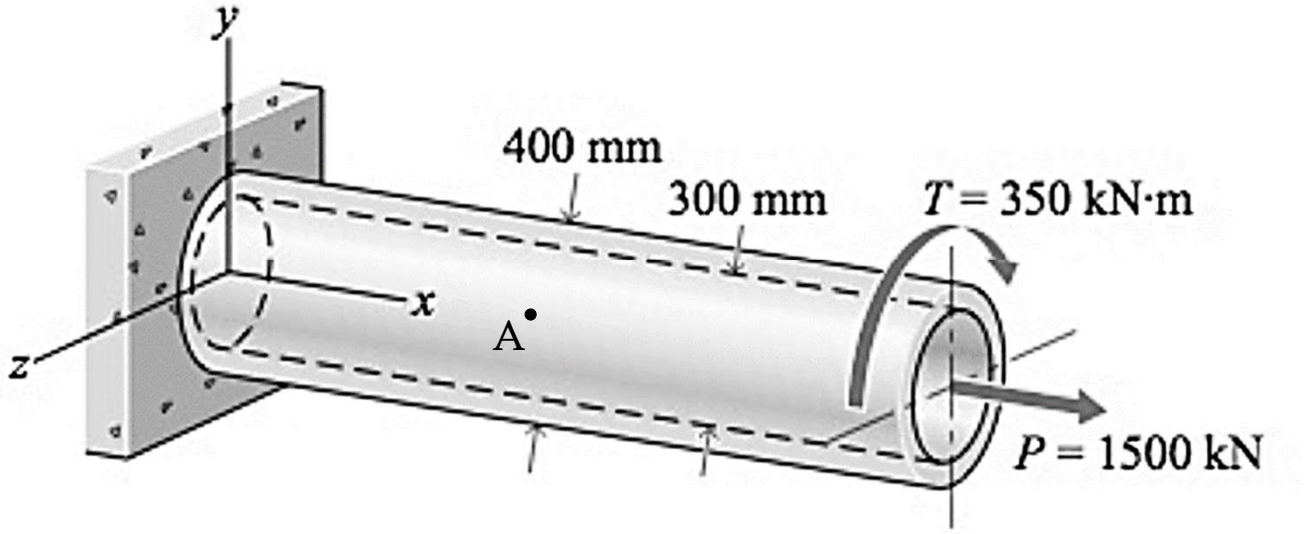
Combined Loading

Example

The solid 100-mm-diameter shaft shown below is subjected to an axial compressive force of P = 200 kN and a torque of T = 30 kNm. For point A on the outside surface of the shaft, determine

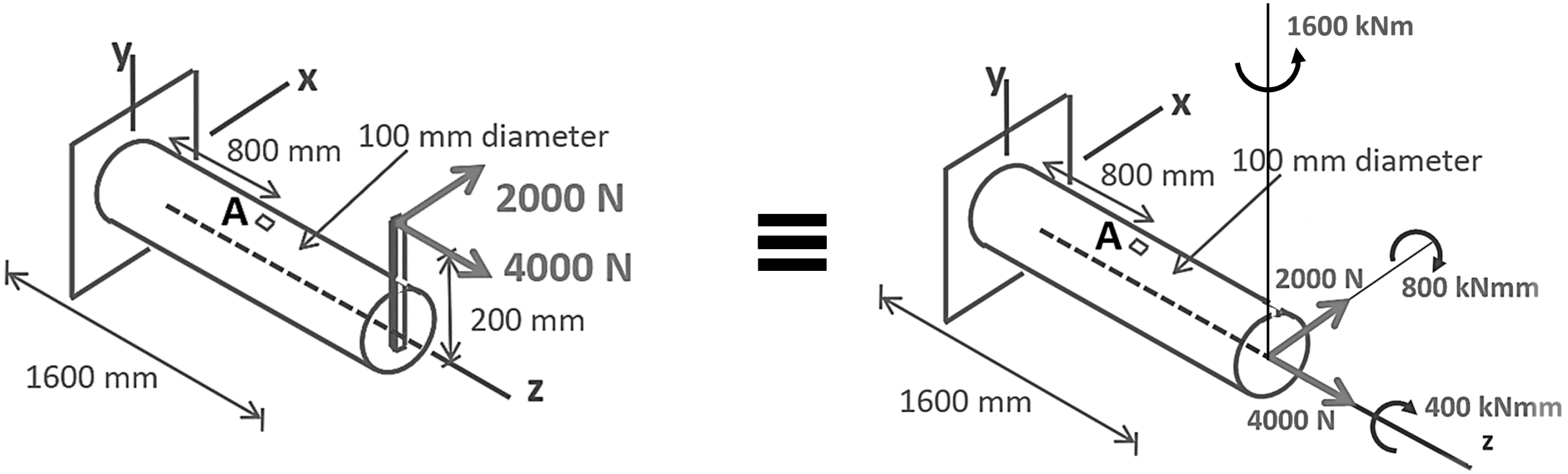
(a) The x- and y-components of stress.

(b) The principal stresses and the maximum shearing stress at the point A, on the outermost part of the shaft.



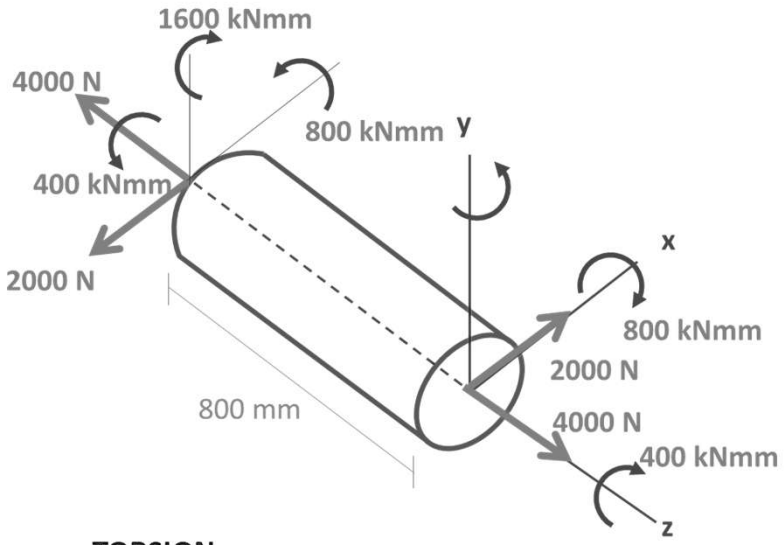
Combined Loading

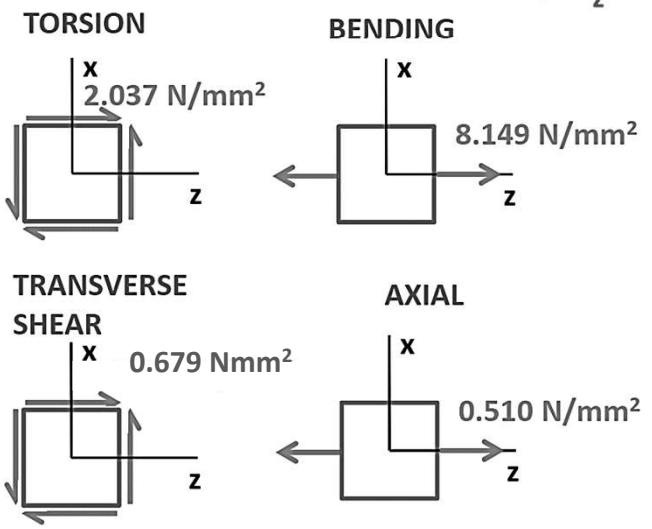
Example



Take point A to at (0,50,800)

Combined Loading





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